Heart Failure Mortality Predictions

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Problem Definition

- Predicting short-term heart failure mortality risk from routine measurements
- Doctors look at routine measurements and make clinical decisions
- Make a model that can alert a care team of mortality risk during follow-up period from admission measurements
  - The care team can react faster to any missed risk indicators

Problem	**Model**	**Background**
Clinic staff resources limited	Neural network maps clinical measurements, ‘x’, to a risk score, P(death | x) in the next 285 days	Chicco & Jurman (2020) found that ejection fraction and serum creatinine measurements predicted survival with random forests.

Dataset

Heart Failure Clinical Records UCI

- 299 patients
- 12 clinical features
- Binary outcome [DEATH_EVENT]

Data Problems

- Small n (299)
- Class imbalance (96 of 299 died during follow-up)
- Feature scaling (i.e. platelet count 100,000s vs. serum creatinine <9)

Deployment Environment

- Routine labs and measurements are recorded on admittance for heart failure pt
- The care team looks at each measurement to make clinical decisions
- The model estimates risk of death during follow-up period
- If the probability of a death event meets the set decision threshold, then the care team is alerted to the patient
- The care team is able to review the measurements and take action, potentially saving the patient's life

Deployment Problems

- False Negative prediction much worse than a False Positive
- Question of demographic generalizability(data from one hospital in 2015)

Base Model

- 12 standardized inputs → 32 hidden(ReLU) → 16 hidden(ReLU) → 1 output
- Sigmoid of output to give probability of death event
- Binary Cross Entropy loss compare output probability with ground truth
- Backpropagate to compute gradients, Adam updates weights
- Minimum of 20 epochs, max of 300
  - Early stopping with patience 20
- Stratified 5-fold cross-validation to give 5 test scores of mean and standard deviation
- No L2 or dropout
- No class balancing

Experiment Overview

Experiment 1 – Class Imbalance

- Does oversampling minority class improve recall on deaths?
- Recall = deaths caught / total deaths

Experiment 2 – Cost Minimization

- Which decision threshold minimizes cost?
- Missed death much worse than false alarm

Experiment 3 – Small n overfitting

- Does regularizing improve the generalizability?
- Training loss might go to 0 while validation loss is high
- Try L2 and dropout

Experiment 4 – Feature Scaling

- How does standardizing inputs improve convergence speed and validation loss over raw inputs?



Experiment 2: Cost Evaluation

Research Question/Reasoning

- Where should the decision threshold be?
- Missing a death prediction is much more costly than a false alarm

For example:

- Correct Decisions = 0 unit cost
- False Positive = 1 unit cost
- False Negative = 20 unit cost (missed death prediction)
  - Repeat for different cost ratios

Experiment Design

1. Train model with plain BCE in each CV fold (239/299 total patients)
2. On a validation split of training folds, try thresholds from 0.01 to 0.95 increment by 0.01
   1. Keep threshold that minimizes expected cost
   2. Average Cost = [20xFNs + 1xFPs]/N
3. Calculate theoretical optimal decision threshold based on cost ratios
   1. Theoretical optimum = 1/(1+20) = 0.048
4. Test base, theoretical, and decision thresholds on the held out test fold

Thank you!



References

- Chicco, D., Jurman, G. Machine learning can predict survival of patients with heart failure from serum creatinine and ejection fraction alone. *BMC Med Inform Decis Mak* **20**, 16 (2020). https://doi.org/10.1186/s12911-020-1023-5